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A study of multidimensional fifth-order WENO method for genuinely two-dimensional Riemann solver

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ABSTRACT

In this paper we conduct a research on the multidimensional fifth-order reconstruction method for Balsara's genuinely two-dimensional (2D) HLLE Riemann solver. This solver possesses the multidimensional effect successfully by solving the 2D Riemann problem at the cell vertexes. However, it is difficult to be applied to higher-order reconstruction technology. In order to meet the requirement of the reconstruction of the variables at the vertexes, we construct the general WENO reconstruction formula to complete the reconstruction procedure along different spatial directions in turns. The Gaussian quadrature is employed to achieve the fifth-order accuracy of the numerical flux integration at the interface. Typical Simpson rule in Balsara's 2D HLLE solver is no longer used, and each Gaussian point is regarded as an independent 2D Riemann problem to obtain the 2D Riemann flux directly. Due to the inevitable numerical overshoots in the process of the fifth-order reconstruction, an accuracy-preserving limiter is introduced to enhance the computational stability, which could maintain the numerical accuracy in the smooth regions and suppress the overshoots near discontinuities. Solution accuracy tests show that the multidimensional WENO method could achieve the fifth-order accuracy. Several solution quality tests are also presented, which indicate that the method proposed in this study performs well in all test cases and can capture discontinuities and complex flow structures accurately and efficiently.

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1. Introduction

In the last few decades, it has always been a hot spot to accurately and efficiently simulate complex multidimensional flows in the community of the Computational Fluid Dynamics (CFD). In general, the spatial discretization method, which mainly includes the flux scheme (also known as Riemann solvers) and the reconstruction method, is regarded as a key factor affecting the accuracy and efficiency of CFD.

In terms of the flux scheme, many researchers have conducted a study on the approximate Riemann solvers and a host of typical methods have sprung up since the introduction of Godunov's method based on exact Riemann solution [1]. Typical Riemann solvers include Flux Difference Splitting (FDS) type schemes [2,3], Flux Vector Splitting (FVS) type schemes [4,5], HLL type schemes [6–8] and AUSM type schemes [9,10]. All of these methods are derived based on the idea of dimensional splitting and are generally used to solve one-dimensional (1D) Riemann flux, so they are called 1D Riemann solvers. However, these solvers lose the calculation accuracy and stability in simulating complex multidimensional

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flows because they omit the flow information carried by the transverse waves along the cell interfaces [11]. Therefore, many scholars paid their attentions to the construction of the genuinely multidimensional Riemann solver. For example, Collela [12], LeVeque [13] carried out the work by building some level of multidimensionality out of the 1D Riemann solvers. Wendroff proposed the multidimensional HLLE scheme [14]. Abgrall [15,16], Fey [17,18] and Brio [19] conducted the multidimensional extensions of Roe's linearized Riemann solvers. Generally, these multidimensional Riemann solvers are indeed able to meet the design requirements of increasing the CFL number and improving the multidimensional properties of numerical solutions, but their complex construction processes result in much higher computational costs than those of the 1D Riemann solvers.

In order to overcome these disadvantages aforementioned, Balsara proposed a genuinely multidimensional Riemann solver by introducing the vertex-based framework [20]. By solving the two-dimensional (2D) Riemann flux at the corners of grid cells, this solver can realize the multidimensional effect of the scheme in a simple and efficient way. Based upon such work, Balsara conducted a series of researches to make further achievements [21–24]. Also, many other improved genuinely multidimensional Riemann solver were proposed. For example, Vides et al. extended the HLL scheme to two-dimension by a simple method that can simplify numerical implementation[25,26]. Capdeville et al. developed a four-state bi-dimensional HLL solver and a simplified central WENO procedure on unstructured triangular meshes[27]. Mandal and Sharma proposed a 2D HLL Riemann solver by combining the HLL-CPS method with the Zha-Bilgen splitting method [28]. Hu constructed a genuinely multidimensional GM-TV-HLL scheme based on the TV splitting method [11]. Qu expended the genuinely multidimensional Riemann solver to the curvilinear coordinate system [29,30], and constructed a novel Riemann solver for complex viscous flows [31]. In general, these multidimensional Riemannian solvers are with higher accuracy, permit a larger time step, and are also easy to be implemented.

However, all these Riemann solvers mentioned above are usually only of first-order accuracy, and their accuracy needs to be improved. For the spatial discretization technology based on the finite volume (FV) method, the higher-order reconstruction method is a key method to improve the calculation accuracy. The reconstruction procedure of the typical 1D Riemann solver in structured grid is relatively simple, which can be decoupled along the spatial directions, so that only one-dimensional reconstruction needs to be concerned. van Leer first constructed the second-order upwind MUSCL (Monotone Upstream-centered Schemes for Conservative Laws) scheme to extend the first-order Godunov scheme to the second-order, and eliminated oscillations near discontinuities by introducing limiters [32]. Then Harten proposed the conception of TVD (Total Variation Diminishing) and developed a series of TVD schemes [33]. However, this scheme can only achieve second-order accuracy at best and degrades to the first-order accuracy at local extrema in the smooth regions. To solve this problem, the TVB (Total Variation Bounded) concept [34] and the ENO (Essentially Non-oscillatory) schemes were proposed [35]. Subsequently, WENO (Weighted-ENO) schemes were developed to improve the computational efficiency of the ENO schemes [36]. The classical WENO scheme proposed by Jiang and Shu can achieve $(2r-1)$th-order accuracy [37]. For the construction methods of the nonlinear weights, people developed a series of revised WENO schemes to improve the numerical accuracy [38,39].

For these, it is an attractive choice to combine the multidimensional Riemann solver with the higher-order reconstruction method, which can improve the resolution of wave-system structures, increase the maximum allowed CFL number, and reduce the calculation errors and costs. However, due to the vertex-framework adopted by the multidimensional Riemann solver, classical higher-order reconstruction method aforementioned cannot meet the requirements of the variable reconstruction at the cell vertexes. And there is little research on the higher-order reconstruction methods for the genuinely multidimensional Riemann solver besides Balsara's work by adopting the ADER-WENO schemes [22].

In this study, the multidimensional fifth-order WENO method is proposed for Balsara's genuinely 2D Riemann solver. The basic idea is to construct the WENO method that decoupled along the spatial direction, and to obtain the reconstructed value around the specified point as the input states to solve the 2D Riemann problem. Then the numerical flux of the cell interface is simulated by employing the Gaussian quadrature role. Also, an accuracy-preserving limiter, which works only near the discontinuities to depress the overshoots and maintain the accuracy in the smooth regions, is adopted in the WENO reconstruction procedure.

This paper is organized as follows. In Section 2, the governing equations are briefly described. In Section 3, the genuinely multidimensional Riemann solver proposed by Balsara is introduced. Section 4 describes the detailed construction procedure of the multidimensional fifth-order WENO method and the accuracy-preserving limiter. Several test cases are conducted to estimate the properties of the fifth-order scheme in Section 5. Finally, the conclusions are given in Section 6.

2. Governing equations

The two-dimensional Euler equations can be generally written in the following differential form:

$$\frac{\partial \boldsymbol{q}}{\partial t}+\frac{\partial \boldsymbol{f}(\boldsymbol{q})}{\partial x}+\frac{\partial \boldsymbol{g}(\boldsymbol{q})}{\partial y}=0 \tag{1}$$

where $\boldsymbol{q}$ is the vector of conserved variables, $\boldsymbol{f}$ and $\boldsymbol{g}$ are flux vectors in the x and y directions as follows:

$$\boldsymbol{q}=\begin{bmatrix}\rho\\ \rho u\\ \rho v\\ E\end{bmatrix},\quad \boldsymbol{f}(\boldsymbol{q})=\begin{bmatrix}\rho u\\ \rho u^2+p\\ \rho uv\\ u(E+p)\end{bmatrix},\qquad \boldsymbol{g}(\boldsymbol{q})=\begin{bmatrix}\rho v\\ \rho uv\\ \rho v^2+p\\ v(E+p)\end{bmatrix} \tag{2}$$

Here, ρ is the density, p is the static pressure, u and v are the velocity components in the x and y directions respectively, and E is the total energy.

For Cartesian grids, define the cell $V_{i,j}=\left[x_{i-1/2},x_{i+1/2}\right]\times\left[y_{j-1/2},y_{j+1/2}\right]$, and the grid spacings in the x and y direction are denoted by Δx and Δy. The integration form of Eq. (1) over the cell $V_{i,j}$ can be written as follows:

$$\frac{d\bar{\boldsymbol{q}}_{i,j}}{dt}=-\frac{1}{\Delta x}\left(\boldsymbol{f}_{i+1/2,j}-\boldsymbol{f}_{i-1/2,j}\right)-\frac{1}{\Delta y}\left(\boldsymbol{g}_{i,j+1/2}-\boldsymbol{g}_{i,j-1/2}\right) \tag{3}$$

where $\bar{\boldsymbol{q}}_{i,j}$ is the mean value of the cell $V_{i,j}$:

$$\bar{\boldsymbol{q}}_{i,j}=\frac{1}{\Delta x\Delta y}\iint\limits_{V_{i,j}}\boldsymbol{q}(x,y)\,dxdy \tag{4}$$

$\boldsymbol{f}_{i+1/2,j}$ and $\boldsymbol{g}_{i,j+1/2}$ are the numerical flux at the interface, and can be solved by the Riemann solvers. For typical 1D Riemann solver, $\boldsymbol{f}_{i+1/2,j}$ is obtained by

$$\boldsymbol{f}_{i+1/2,j}=\boldsymbol{f}\left(\boldsymbol{q}^L_{i+1/2},\boldsymbol{q}^R_{i+1/2}\right) \tag{5}$$

The $\boldsymbol{q}^L_{i+1/2},\boldsymbol{q}^R_{i+1/2}$ can be seen as the values on both sides of the interface, which should be reconstructed by higher-order interpolation if higher accuracy is wanted. Notice that for multidimensional Riemann solver, more values around the interface are needed, and we will expatiate it in the next section.

In this paper, the third-order TVD Runge-Kutta integration method is used for the time evolution to obtain the updated variable $\boldsymbol{q}^{n+1}$:

$$\begin{aligned}\boldsymbol{q}^{(1)}&=\boldsymbol{q}^n+\Delta tL\left(\boldsymbol{q}^n\right)\\ \boldsymbol{q}^{(2)}&=\frac{3}{4}\boldsymbol{q}^n+\frac{1}{4}\boldsymbol{q}^{(1)}+\frac{1}{4}\Delta tL\left(\boldsymbol{q}^{(1)}\right)\\ \boldsymbol{q}^{n+1}&=\frac{1}{3}\boldsymbol{q}^n+\frac{2}{3}\boldsymbol{q}^{(2)}+\frac{2}{3}\Delta tL\left(\boldsymbol{q}^{(2)}\right)\end{aligned} \tag{6}$$

where superscript n denotes the time level, and L is the residual obtained by the right item of Eq. (3).

3. The genuinely two-dimensional Riemann solver by Balsara

According to the analysis in Section 1, Balsara's genuinely 2D HLLE Riemann solver is chosen in this paper because of its simple closed-form and high efficiency. In this part, we will introduce the construction procedure of this solver briefly, more details can be found in Ref. [20]. In consideration of clear expression, the single component is used instead of the vector variables (say q instead of $\boldsymbol{q}$) in the governing equations in the following sections.

As shown in Fig. 1, a vertex framework is used to obtain the numerical flux at the interface, which is assembled with the 1D Riemann flux solved at the midpoint and the 2D Riemann flux solved at the cell vertexes or corner points (such as C1∼C4). The former is solved by typical 1D HLLE scheme, while the latter is solved by the 2D HLLE Riemann solver that derived based on the 2D wave propagation model. This model defines a 2D Riemann problem around the vertex and takes both the influence of transverse wave and orthogonal wave into consideration. Unlike the 1D Riemann solver, the 2D solver requires four additional input states (RU, LU, LD, RD) at each vertex to obtain the numerical flux. For example, the 2D Riemann flux at vertex C1 is obtained by

$$\begin{aligned}f^c_{i+1/2,j+1/2}=&\frac{f_{LU}S^c_RS^c_U+f_{RD}S^c_LS^c_D-f_{LD}S^c_RS^c_D-f_{RU}S^c_LS^c_U}{\left(S^c_R-S^c_L\right)\left(S^c_U-S^c_D\right)}\\ &-2\left[\frac{S^c_RS^c_L}{\left(S^c_R-S^c_L\right)\left(S^c_U-S^c_D\right)}\right](g_{RU}-g_{LU}+g_{LD}-g_{RD})\\ &+\left[\frac{S^c_RS^c_L}{\left(S^c_R-S^c_L\right)\left(S^c_U-S^c_D\right)}\right]\left[S^c_U\left(q_{RU}-q_{LU}\right)-S^c_D\left(q_{RD}-q_{LD}\right)\right]\end{aligned} \tag{7}$$

where the superscript "c" means the values related to the vertexes of the interface. S^c_L, S^c_R, S^c_U, S^c_D are the largest wave speeds towards four directions.

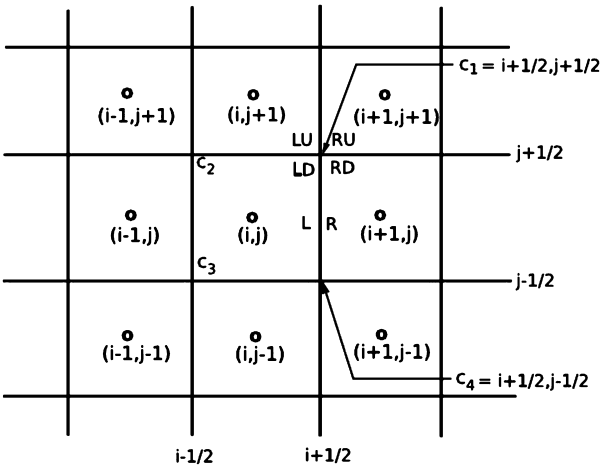


Fig. 1. Two-dimensional computational mesh [40].

The numerical fluxes at the vertex C2, C3 and C4, which are denoted by $f^c_{i+1/2,j-1/2}$, $f^c_{i-1/2,j+1/2}$ and $f^c_{i-1/2,j-1/2}$ respectively, can be solved in the same way. The final numerical flux at the interface $i+1/2$ is assembled with a Simpson rule:

$$f_{i+1/2,j} = \frac{1}{6} f^c_{i+1/2,j+1/2} + \frac{4}{6} f^m_{i+1/2,j} + \frac{1}{6} f^c_{i+1/2,j-1/2} \tag{8}$$

where $f^m_{i+1/2,j}$ represents 1D Riemann flux at the midpoint of the interface.

Now come to the question of how to obtain the higher-order reconstructed values of q_{RU}, q_{LU}, q_{RD}, q_{LD} to achieve higher accuracy. Traditional high-order reconstruction methods suitable for structured grid are generally derived in the 1D case to solve $q^L_{i+1/2}$, $q^R_{i+1/2}$. However, these 1D reconstruction methods face with the difficulty to complete the reconstruction at the cell vertexes or some arbitrary points, which prevents them from meeting the requirements of variable reconstruction for the genuinely multidimensional Riemann solver. The reconstruction methods suitable for unstructured grid can solve this problem easily. For example, we can utilize the unstructured HWENO methods proposed by Qiu and Shu [41] to solve the reconstructed values at the cell vertexes. However, these kinds of reconstruction methods need multidimensional interpolation stencils with a lot of cells, and the number of cells and computational costs increase rapidly with the increase of the accuracy order. Thus, a deep study of multidimensional higher-order scheme suitable for the multidimensional Riemann solvers is necessary. In the next section, we will conduct research on the fifth-order WENO reconstruction methods which are decoupled along the spatial directions, aiming to obtain the higher-order reconstructed values for the multidimensional Riemann solver in a simple and effective way.

4. Fifth-order reconstruction methods

4.1. General fifth-order WENO reconstruction method

In this section, the general fifth-order WENO reconstruction method is introduced to obtain the reconstructed values at any position of the grid cells [42]. Suppose there is a one-dimensional uniform grid $I_i = \left[x_{i-1/2}, x_{i+1/2}\right]$ with the grid spacing Δx, and an unknown function distribution $f(x)$ exists on the grid cell. $\bar{f}_i$ is known as the average of $f(x)$ inside I_i. Five adjacent cells $\{I_{i-2}, I_{i-1}, I_i, I_{i+1}, I_{i+2}\}$ that centered with cell I_i are used as the interpolation stencil to construct the quartic reconstruction polynomial as the fifth-order approximation of $f(x)$ at any interpolation point in the cell I_i, and the quartic reconstruction polynomial is denoted as $Q(x)$.

In general WENO method, the quartic polynomial $Q(x)$ is approximated by a combination of three quadratic polynomials to achieve the highest accuracy in the smooth regions and suppress oscillations near the discontinuous regions. Three small stencils divided from the five-cell stencil are utilized to construct the quadratic polynomials, and the small stencils are given by

$$S_0 = \{I_{i-2}, I_{i-1}, I_i\}, \qquad S_1 = \{I_{i-1}, I_i, I_{i+1}\}, \qquad S_2 = \{I_i, I_{i+1}, I_{i+2}\} \tag{9}$$

The quadratic polynomials on respective stencils, which are denoted by $p_j(x)$, $j = 0, 1, 2$ with the form $p_j(x) = a_j x^2 + b_j x + c_j$, should satisfy the following conditions:

$$\frac{1}{\Delta x} \int_{I_k} p_0(x)\, dx = \bar{f}_k, \quad k = i-2, i-1, i$$

$$\frac{1}{\Delta x}\int_{I_k} p_1(x)\,dx = \bar{f}_k, \quad k = i-1, i, i+1$$
$$\frac{1}{\Delta x}\int_{I_k} p_2(x)\,dx = \bar{f}_k, \quad k = i, i+1, i+2 \tag{10}$$

The coefficients a_j, b_j, c_j can be solved through a linear system of equations by substituting $p_j(x) = a_j x^2 + b_j x + c_j$ into Eq. (10). For convenience of subsequent expression and calculation, we use local coordinates instead of global coordinates. The local coordinate of interpolation point is denoted by $x^G = x_{i-1/2} + \alpha\Delta x$, where $\alpha \in [0,1]$ in the cell $V_{i,j}$. Denote the new quadratic polynomials by $\hat{p}_j(\alpha)$, $j = 0, 1, 2$, and the third-order approximations at the point x^G obtained after the transformation are as follows:

$$\hat{p}_0(\alpha) = \frac{1}{6}\left(\left(3\alpha^2 - 1\right)\bar{f}_{i-2} + \left(-6\alpha^2 - 6\alpha + 5\right)\bar{f}_{i-1} + \left(3\alpha^2 + 6\alpha + 2\right)\bar{f}_i\right)$$
$$\hat{p}_1(\alpha) = \frac{1}{6}\left(\left(3\alpha^2 - 6\alpha + 2\right)\bar{f}_{i-1} + \left(-6\alpha^2 + 6\alpha + 5\right)\bar{f}_i + \left(3\alpha^2 - 1\right)\bar{f}_{i+1}\right)$$
$$\hat{p}_2(\alpha) = \frac{1}{6}\left(\left(3\alpha^2 - 12\alpha + 11\right)\bar{f}_i + \left(-6\alpha^2 + 18\alpha - 7\right)\bar{f}_{i+1} + \left(3\alpha^2 - 6\alpha + 2\right)\bar{f}_{i+2}\right) \tag{11}$$

Similarly, the quartic interpolation polynomial $Q(x)$, which constructed by all the five interpolated points, should satisfy the following conditions:

$$\frac{1}{\Delta x}\int_{I_j} Q(x)\,dx = \bar{f}_j, \quad j = i-2, i-1, i, i+1, i+2 \tag{12}$$

And the new polynomial after converting to local coordinates is denoted by $\hat{Q}(\alpha)$.

In smooth regions, to achieve the highest order of accuracy, the combination of three quadratic polynomials should equal to the quartic polynomial.

$$\hat{Q}(\alpha) = \gamma_0\hat{p}_0(\alpha) + \gamma_1\hat{p}_1(\alpha) + \gamma_2\hat{p}_2(\alpha) \tag{13}$$

Where the γ_0, γ_1, γ_2 could bear the name ideal weights. They can be solved by simple algebraic manipulation:

$$\gamma_0 = \frac{5\alpha^4 - 20\alpha^3 + 15\alpha^2 + 10\alpha - 6}{60\alpha^2 - 20}$$
$$\gamma_1 = \frac{-30\alpha^6 + 90\alpha^5 + 55\alpha^4 - 260\alpha^3 + 81\alpha^2 + 64\alpha - 24}{20\left(3\alpha^2 - 1\right)\left(3\alpha^2 - 6\alpha + 2\right)}$$
$$\gamma_2 = \frac{5\alpha^4 - 15\alpha^2 + 4}{60\alpha^2 - 120\alpha + 40} \tag{14}$$

However, in regions near discontinuities, the stencils which contain discontinuities should be assigned a nearly zero weight to satisfy the essentially non-oscillatory property. There must be some methods to construct new non-oscillatory weights based on the ideal weights. To meet this requirement, the smooth indicator is introduced to represent the smoothness of interpolation polynomials. According to Jiang and Shu's idea [37], the smooth indicator is calculated based upon minimizing the L^2 norm of the reconstruction polynomials. In this work, the smooth indicator IS_k of each small stencil S_k under the local coordinates is defined as follows:

$$IS_k = \sum_{l=1}^{r-1}\int_0^1 \left(\hat{p}_k^{(l)}(\alpha)\right)^2 d\alpha, \quad k = 0, 1, 2 \tag{15}$$

Where $\hat{p}_k^{(l)}(\alpha)$ is the lth-derivative of $\hat{p}_k(\alpha)$. The right-hand side of Eq. (15) is the sum of the L^2 norm of all the derivatives of the reconstruction polynomial $\hat{p}_k(\alpha)$ over the interval $\alpha \in [0,1]$. So for $r = 3$, Eq. (15) gives

$$IS_0 = \frac{1}{4}\left(\bar{f}_{i-2} - 4\bar{f}_{i-1} + 3\bar{f}_i\right)^2 + \frac{13}{12}\left(\bar{f}_{i-2} - 2\bar{f}_{i-1} + \bar{f}_i\right)^2$$
$$IS_1 = \frac{1}{4}\left(\bar{f}_{i-1} - \bar{f}_{i+1}\right)^2 + \frac{13}{12}\left(\bar{f}_{i-1} - 2\bar{f}_i + \bar{f}_{i+1}\right)^2$$
$$IS_2 = \frac{1}{4}\left(3\bar{f}_i - 4\bar{f}_{i+1} + 3\bar{f}_{i+2}\right)^2 + \frac{13}{12}\left(\bar{f}_i - 2\bar{f}_{i+1} + \bar{f}_{i+2}\right)^2 \tag{16}$$

With smooth indicators, the non-oscillatory weights can be solved by Borges's weighting function [43]:

$$\begin{aligned} IS_k^z &= \frac{IS_k+\varepsilon}{IS_k+\tau_5+\varepsilon}, \quad k=0,1,2 \\ \omega_k^z &= \frac{\alpha_k^z}{\alpha_0^z+\alpha_1^z+\alpha_2^z}, \qquad \alpha_k^z = \frac{\gamma_k}{IS_k^z} = \gamma_k\left(1+\left(\frac{\tau_5}{IS_k+\varepsilon}\right)^q\right) \\ \tau_5 &= |IS_0 - IS_2| \end{aligned} \tag{17}$$

Final reconstruction polynomial is

$$\hat{Q}^z(\alpha) = \omega_0^z \hat{p}_0(\alpha) + \omega_1^z \hat{p}_1(\alpha) + \omega_2^z \hat{p}_2(\alpha) \tag{18}$$

From this polynomial, we can obtain the reconstructed values at any position of the grid cells. For example, we can get the values at the cell interface $i+1/2$ when $\alpha=1$. However, one may get the negative ideal weights when α gets some value, such as $\alpha=1/2$, which causes that the reconstruction near the discontinuous region cannot be essentially non-oscillatory. A simple and effective separation method proposed by Shi and Hu [44] is adopted to solve this problem. The detailed implementation process is given below.

If $\min(\gamma_0,\gamma_1,\gamma_2) \geq 0$, no additional processing is required. If $\min(\gamma_0,\gamma_1,\gamma_2) < 0$, the ideal weights should be divided into two parts:

$$\bar{\gamma}_j^+ = \frac{1}{2}\left(\gamma_j + \theta|\gamma_j|\right), \quad \bar{\gamma}_j^- = \bar{\gamma}_j^+ - \gamma_j, \quad j=0,1,2 \tag{19}$$

where $\theta=3$. Then the ideal weights of the positive and negative parts are constructed respectively:

$$\gamma_j^{\pm} = \frac{\bar{\gamma}_j^{\pm}}{\sigma^{\pm}}, \quad \sigma^{\pm} = \sum_{j=0}^{2} \bar{\gamma}_j^{\pm}, \quad j=0,1,2 \tag{20}$$

And we can get two separate polynomials:

$$Q^{\pm}\left(x^G\right) = \sum_{j=0}^{2} \gamma_j^{\pm} p_j\left(x^G\right) \tag{21}$$

which satisfy that

$$Q\left(x^G\right) = \sigma^+ Q^+\left(x^G\right) - \sigma^- Q^-\left(x^G\right) \tag{22}$$

The non-oscillatory weights of the positive and negative parts can also be defined as follows:

$$\omega_j^{\pm} = \frac{\tilde{\omega}_j^{\pm}}{\sum_j \tilde{\omega}_j^{\pm}}, \qquad \tilde{\omega}_j^{\pm} = \frac{\gamma_j^{\pm}}{\left(\varepsilon + IS_j\right)^2} \tag{23}$$

So,

$$Q^{\pm}\left(x^G\right) = \sum_{j=0}^{2} \omega_j^{\pm} p_j\left(x^G\right) \tag{24}$$

Final WENO reconstruction polynomial is assembled by

$$Q\left(x^G\right) = \sigma^+ Q^+\left(x^G\right) - \sigma^- Q^-\left(x^G\right) \tag{25}$$

Through the above separation method, a group of ideal weights with negative values can be divided into two groups of weights with both positive values, so as to avoid the oscillation of numerical solutions.

4.2. Reconstruction procedure for the genuinely multidimensional Riemann solver

4.2.1. Basic idea of multidimensional reconstruction procedure

In this section, the multidimensional WENO scheme is studied based on the general WENO formula derived in Section 4.1. We denoted this multidimensional WENO scheme as MD_WENO_Z5 method due to its non-oscillatory weights' construction method. The spatial discretization form of the Euler equations has been shown in Eq. (3). A three-points Gaussian quadrature is conducted to achieve the fifth-order accuracy for the two-dimensional cases [45]. The flux $f_{i+1/2,j}$ and $g_{i,j+1/2}$ in Eq. (3) are approximately integrated as follows:

$$\begin{cases} f_{i+1/2,j} \approx \sum_{k=1}^{3} \beta_k f\left(q^{LD}_{i+1/2,j_k}, q^{LU}_{i+1/2,j_k}, q^{RD}_{i+1/2,j_k}, q^{RU}_{i+1/2,j_k}\right) \\ g_{i,j+1/2} \approx \sum_{k=1}^{3} \beta_k g\left(q^{LD}_{i_k,j+1/2}, q^{LU}_{i_k,j+1/2}, q^{RD}_{i_k,j+1/2}, q^{RU}_{i_k,j+1/2}\right) \end{cases}, \quad k=1,2,3 \tag{26}$$

where $f\left(q^{LD}_{i+1/2,j_k}, q^{LU}_{i+1/2,j_k}, q^{RD}_{i+1/2,j_k}, q^{RU}_{i+1/2,j_k}\right)$ and $g\left(q^{LD}_{i_k,j+1/2}, q^{LU}_{i_k,j+1/2}, q^{RD}_{i_k,j+1/2}, q^{RU}_{i_k,j+1/2}\right)$ are genuinely 2D Riemann solvers shown in Section 3, β_k is Gauss quadrature coefficient. Significantly, both the 1D and 2D Riemann problems can be solved by the above 2D Riemann solvers. Because when $q^{LU}_{i+1/2,j_2} = q^{LD}_{i+1/2,j_2}$ and $q^{RU}_{i+1/2,j_2} = q^{RD}_{i+1/2,j_2}$, $f\left(q^{LD}_{i+1/2,j_k}, q^{LU}_{i+1/2,j_k}, q^{RD}_{i+1/2,j_k}, q^{RU}_{i+1/2,j_k}\right)$ will degenerate to 1D Riemann solver. Therefore, all the 1D Riemann problems in this work are also solved by the 2D Riemann solvers. The key of multidimensional reconstruction procedure is to reconstruct the input values at the desired position for the multidimensional Riemann solver. In Ref. [45], the author constructed a WENO scheme in 2D Cartesian grids to obtain the reconstructed values at the Gaussian points on both sides of the interface. Based on his idea, we make some changes to obtain the four reconstructed values around the Gaussian points, which are used as input states for genuinely multidimensional Riemann solver.

According to the general fifth-order reconstruction method, the multidimensional reconstruction procedure can be constructed based on dimension splitting. For the input of $f_{i+1/2,j}$ at interface $i+1/2$, the reconstruction procedure can be divided into two steps. Firstly, use the average value ($\bar{q}_{i,j}$) of grid cells to perform one-dimensional fifth-order reconstruction along x-direction to obtain high order of approximations on both sides of the interface $i+1/2$ ($\bar{q}^{L}_{i+1/2,j}$ and $\bar{q}^{R}_{i+1/2,j}$). And then, use the reconstructed values obtained in the first step to perform one-dimensional fifth-order reconstruction along y-direction to obtain four input values ($q^{LD}_{i+1/2,j_k}$, $q^{LU}_{i+1/2,j_k}$, $q^{RD}_{i+1/2,j_k}$, $q^{RU}_{i+1/2,j_k}$) of $f_{i+1/2,j}$ at every Gauss point. The input values at the interface $j+1/2$ can be reconstructed by the same method but in reverse order. Theoretically, the reconstruction procedure satisfies the essentially non-oscillatory property due to the smooth indicators and non-oscillatory weights. But numerical results show that it is not robust enough and still exists overshoots near the discontinuities in simulating complex flows. Therefore, some methods must be introduced to suppress unphysical oscillation without loss of accuracy.

4.2.2. An accuracy-preserving limiter that enhances stability

In this section, an accuracy-preserving limiter is constructed to enhance computational stability. Barth's idea of restriction is adopted and the reconstruction near the discontinuities is restricted to ensure that there is no new extrema [46]. That is, assuming that variables q satisfy $q \in [m, M]$, we hope that the reconstructed values q^{Gauss} at the Gaussian points also satisfy $q^{Gauss} \in [m, M]$. In the x direction, q^{Gauss} include $q^{LD}_{i+1/2,j_k}$, $q^{LU}_{i+1/2,j_k}$, $q^{RD}_{i+1/2,j_k}$, $q^{RU}_{i+1/2,j_k}$, $(k=1,2,3)$. Generally, q^{Gauss} obtained from the reconstruction procedure in Section 4.2.1 cannot always satisfy $q^{Gauss} \in [m, M]$, thus some extra processing is required.

Suppose that the fifth-order approximation polynomial in cell $V_{i,j}$ is denoted as $p_{i,j}(x, y)$. A form of limiter similar to that proposed by Zhang [47] is imposed for $p_{i,j}(x, y)$ near the discontinuities, and the restricted polynomial is denoted as $\tilde{p}_{i,j}(x, y)$. The process of restriction is as follows:

$$\tilde{p}_{i,j}(x, y) = \theta\left(p_{i,j}(x, y) - \bar{q}_{i,j}\right) + \bar{q}_{i,j}, \qquad \theta = \min\left\{\left|\frac{M - \bar{q}_{i,j}}{M_V - \bar{q}_{i,j}}\right|, \left|\frac{m - \bar{q}_{i,j}}{m_V - \bar{q}_{i,j}}\right|, 1\right\} \tag{27}$$

where

$$M_V = \max_{x,y \in V_{i,j}} p_{i,j}(x, y), \quad m_V = \min_{x,y \in V_{i,j}} p_{i,j}(x, y), \quad M = \max\left(\bar{q}_{i,j}, \bar{q}_{neighbours}\right), \quad m = \min\left(\bar{q}_{i,j}, \bar{q}_{neighbours}\right).$$

Obviously, this limiter satisfies conservativity and ensures the reconstructed values not exceed the maximum and minimum of the neighboring values. Zhang has proved that this method could preserve accuracy in the smooth regions when M_V and m_V are the fifth-order approximation of M and m respectively [47]. However, the values of M_V and m_V are usually hard to determine, and we estimate them by

$$M_V = \max_k\left(q^{LD}_{i+1/2,j_k}, q^{LU}_{i+1/2,j_k}, q^{RD}_{i+1/2,j_k}, q^{RU}_{i+1/2,j_k}\right), \qquad m_V = \min_k\left(q^{LD}_{i+1/2,j_k}, q^{LU}_{i+1/2,j_k}, q^{RD}_{i+1/2,j_k}, q^{RU}_{i+1/2,j_k}\right) \tag{28}$$

where the reconstructed values at Gaussian points are used. At this point, M_V is always not the fifth-order approximation of M, which causes the accuracy degrades at the extrema in the smooth regions. In order to preserve the accuracy in the smooth regions, a discontinuity-detector is adopted to modify this limiter and ensure it works near the discontinuities only.

In this section, the conception of standard deviation coefficient [48] is used to construct the discontinuity-detector. A nine-cell stencil shown in Fig. 2 is chosen as the discontinuity-detecting region. And the reconstruction polynomials should

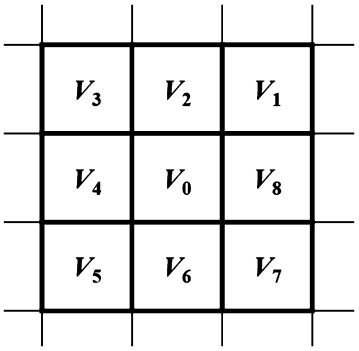


Fig. 2. The discontinuity-detecting region.

be restricted if discontinuity exists in this region. The solution value in each cell is denoted as q_i, $i = 1, 2, \cdots, 9$, which is used to obtain the mean value $\bar{q}$ and the standard deviation σ of the stencil as follows:

$$\bar{q} = \frac{1}{n}\sum_{i=1}^{n} q_i, \qquad \sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(q_i - \bar{q})^2} \tag{29}$$

The standard deviation coefficient is defined as the ratio of the standard deviation and the mean value in the detecting region, and it represents a normalized measure of the degree of data dispersion. That is, the coefficient is small if the stencil is smooth, and it gets much larger if the stencil contains discontinuity. Besides, the standard deviation coefficient in the detecting region may vary a lot as the mesh size h changes. So, we combine the standard deviation coefficient with mesh size h to define the discontinuity-detector by

$$Detector = \frac{\sigma}{h\bar{q}} \tag{30}$$

It is easy to estimate a region whether it contains the discontinuity or not through the following relationship:

$$\begin{cases} Detector < (Detector)_0 & \text{in smooth region} \\ Detector \geq (Detector)_0 & \text{in discontinuous region} \end{cases} \tag{31}$$

Test cases show that $(Detector)_0 = 5$ can always obtain the good results.

Then, we define a new θ as:

$$\tilde{\theta} = \begin{cases} 1 & Detector \leq 5 \\ \min\left\{\left|\frac{M-\bar{q}_{i,j}}{M_V-\bar{q}_{i,j}}\right|, \left|\frac{m-\bar{q}_{i,j}}{m_V-\bar{q}_{i,j}}\right|, 1\right\} & Detector > 5 \end{cases} \tag{32}$$

And the modified accuracy-preserving limiter can be written as

$$\tilde{p}_{i,j}(x, y) = \tilde{\theta}\left(p_{i,j}(x, y) - \bar{q}_{i,j}\right) + \bar{q}_{i,j} \tag{33}$$

4.2.3. The detailed reconstruction procedure with limiter

Now come to the detailed application of the fifth-order reconstruction procedure with the accuracy-preserving limiter in the multidimensional Riemann solver. As shown in Fig. 3, to facilitate the description of interpolation templates, it's necessary to define some interpolation cells. Firstly, define the grid cell that has mean value $\bar{q}_{i,j}$ as $\boldsymbol{V_{i,j}}$. And then, assume that there are two one-dimensional cells on the left and right of the interface $i + 1/2$. They have mean values $\bar{q}^L_{i+1/2,j}$ and $\bar{q}^R_{i+1/2,j}$ respectively and they will be called $\boldsymbol{W^L_{i+1/2,j}}$ and $\boldsymbol{W^R_{i+1/2,j}}$ in the following contents. Take the flux at the interface $i + 1/2(f_{i+1/2,j})$ as an example, the two-dimensional reconstruction procedure can be described as follows:

(1) Use $\bar{q}_{i,j}$ (the solution average of two-dimensional grid cell) to perform the one-dimensional reconstruction in the x-direction according to Eq. (25), and obtain the high-order approximation of q at the cell interface $i + 1/2$. The two values, which are denoted as $\bar{q}^L_{i+1/2,j}$ and $\bar{q}^R_{i+1/2,j}$, can be reconstructed based on the stencil centered with $\boldsymbol{V_{i,j}}$ and $\boldsymbol{V_{i+1,j}}$ respectively. Note that these two values can be used for 1D Riemann solver directly, but they are not enough for genuinely 2D Riemann solvers.

(2) Use $\bar{q}^L_{i+1/2,j}$ and $\bar{q}^R_{i+1/2,j}$ which are obtained in step (1) to complete the one-dimensional reconstruction in the y-direction according to Eq. (25) to obtain the high-order approximation at the Gaussian point y_{j_k}. For the fifth-order scheme, three Gaussian points (G1∼G3 in Fig. 3) are needed. The corresponding coordinates are $y_{j_1} = y_j + \sqrt{15}/10\Delta y$, $y_{j_2} = y_j$ and $y_{j_3} = y_j - \sqrt{15}/10\Delta y$. Among them, the flux at G1 and G3 is seen as a 2D Riemann problem, and the flux

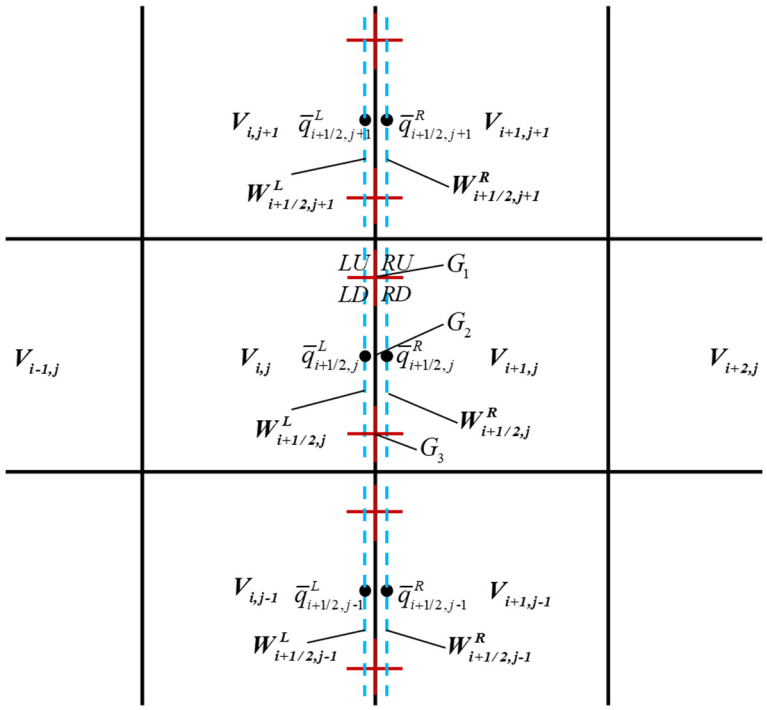


Fig. 3. The solution procedure of multidimensional Riemann solver at the interface.

at G2 is an 1D Riemann problem. As mentioned in section 4.2.1, four reconstructed values are needed for both 1D and 2D Riemann problems. The reconstructed values at G2 have been solved in step (1), $\bar{q}^{LU}_{i+1/2,j_2} = \bar{q}^{LD}_{i+1/2,j_2} = \bar{q}^{L}_{i+1/2,j}$ and $\bar{q}^{RU}_{i+1/2,j_2} = \bar{q}^{RD}_{i+1/2,j_2} = \bar{q}^{R}_{i+1/2,j}$. For the reconstruction at G1 and G3, $\bar{q}^{LD}_{i+1/2,j_1}$ and $\bar{q}^{LU}_{i+1/2,j_3}$ can be reconstructed based on the stencil centered with $\boldsymbol{W^L_{i+1/2,j}}$, and the values of α are $\alpha_1 = 0.5 + \sqrt{15}/10$, $\alpha_3 = 0.5 - \sqrt{15}/10$. $\bar{q}^{LU}_{i+1/2,j_1}$ and $\bar{q}^{LD}_{i+1/2,j_3}$ can be reconstructed based on the stencil centered with $\boldsymbol{W^L_{i+1/2,j+1}}$ and $\boldsymbol{W^L_{i+1/2,j-1}}$ respectively, and the values of α are $\alpha_1 = -(0.5 - \sqrt{15}/10)$, $\alpha_3 = 1.5 - \sqrt{15}/10$. The values on the right side of the interface $i + 1/2$ can be obtained by the same α but different stencils that constructed by the right cells such as $\boldsymbol{W^R_{i+1/2,j}}$, $\boldsymbol{W^R_{i+1/2,j+1}}$.

(3) From step (2), the reconstructed values at the Gaussian points have been prepared for the multidimensional solver. For the uniformity of equation, the reconstructed values at all the Gaussian points are denoted as $\bar{q}^{LD}_{i+1/2,j_k}$, $\bar{q}^{LU}_{i+1/2,j_k}$, $\bar{q}^{RD}_{i+1/2,j_k}$, $\bar{q}^{RU}_{i+1/2,j_k}$, $k = 1, 2, 3$. And then adopt the accuracy-preserving limiter in Section 4.2.2 to modify the reconstructed values obtained in step (2). For example, the modified reconstructed value of $\bar{q}^{LD}_{i+1/2,j_k}$ is $\tilde{q}^{LD}_{i+1/2,j_k} = \tilde{\theta}\left(\bar{q}^{LD}_{i+1/2,j_k} - \bar{q}_{i,j}\right) + \bar{q}_{i,j}$.

(4) Use the modified reconstructed values as the input state values of the multidimensional Riemann solver, and obtain the 2D Riemann flux at each Gaussian point. It should be noticed that this multidimensional Riemann solver doesn't need to take the form of the Simpson rule, and the Riemann problem at each Gaussian point will be solved by 2D Riemann solver. That is, Eq. (7) is used directly to solve the flux $f\left(\tilde{q}^{LD}_{i+1/2,j_k}, \tilde{q}^{LU}_{i+1/2,j_k}, \tilde{q}^{RD}_{i+1/2,j_k}, \tilde{q}^{RU}_{i+1/2,j_k}\right)$, and the final interface flux $f_{i+1/2,j}$ is obtained by the Gaussian quadrature in Eq. (26). The three Gaussian weights are $\beta_1 = 4/9$, $\beta_2 = 5/18$, $\beta_3 = 4/9$.

The interface flux $g_{i,j+1/2}$ can be obtained at the y direction in the same way. Complete the spatial discretization process and obtain the residual L. Then, the time advance solution is carried out according to Eq. (6) until the flow solution converges or reaches the specified calculation time.

5. Numerical results

In this section, the fifth-order multidimensional WENO method described in Section 4 is implemented to the issue of accuracy and some test problems involving Euler equations. The numerical results are compared with those of the third-order unstructured HWNEO method [41] to demonstrate the effect of the fifth-order reconstruction procedure. Balsara's genuinely 2D HLLE Riemann solver and the accuracy-preserving limiter are used in all test cases if there is no particular emphasis. All computations are conducted based on the Intel Fortran Compiler on a Windows system of Intel(R) Xeon(R) E5-2660 (2.6 GHz) by using a single core.

Table 1
L_1 and L_∞ errors and orders for test1 problem.

Method	Grid	L_1 error	L_1 order	L_∞ error	L_∞ order
HWENO	20×20	1.46E-03	–	2.59E-03	–
	40×40	3.02E-04	2.2699	4.82E-04	2.4265
	80×80	4.79E-05	2.6571	7.55E-05	2.6745
	160×160	6.32E-06	2.9240	9.93E-06	2.9271
	320×320	8.00E-07	2.9816	1.26E-06	2.9822
MD_WENO_Z5	20×20	7.75E-04	–	1.21E-03	–
	40×40	2.42E-05	4.9987	3.80E-05	4.9881
	80×80	7.60E-07	4.9952	1.19E-06	4.9930
	160×160	2.38E-08	4.9982	3.73E-08	4.9979
	320×320	7.43E-10	4.9993	1.17E-09	4.9993

5.1. Solution accuracy tests

5.1.1. Test1 problem

Consider a typical two-dimensional linear problem with the following form:

$$\begin{cases} \dfrac{\partial u}{\partial t} + \dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} = 0 \\ u(0, x, y) = \sin(2\pi (x + y)) \end{cases} \tag{34}$$

The exact solution at time t satisfies $u_{exact}(t, x, y) = \sin(2\pi (x + y - 2t))$. This problem runs on a domain of $[0, 1] \times [0, 1]$ with periodic boundary conditions. The time step is adopted as $\Delta t = h^{5/3}$, where h is the grid size. The different norms of the error are defined as follows:

$$\begin{aligned} L_1 &= \frac{1}{N} \sum_{i}^{N} \left| u_{ij_{num}} - u_{ij_{exact}} \right| \\ L_\infty &= \max \left| u_{ij_{num}} - u_{ij_{exact}} \right| \end{aligned} \tag{35}$$

where $u_{ij_{num}}$ and $u_{ij_{exact}}$ denote the numerical and exact values at cell $V_{i,j}$ respectively, and N denotes the total number of cells. The L_1 and L_∞ errors and orders at time $t = 1.0$ for the HWENO and MD_WENO_Z5 methods are given in Table 1. We can see that both methods achieve their designed accuracy. Under the same mesh size, the absolute error values of the HWENO method are larger than the MD_WENO_Z5 method, which indicates that the MD_WENO_Z5 method obtains the lower dissipation.

5.1.2. Test2 problem

A two-dimensional accuracy test involving Euler equations is considered in this section. The computational domain is $[0, 2] \times [0, 2]$. The initial conditions are given by

$$(\rho, u, v, p) = (1.0 + 0.2 \sin(\pi (x + y)), 1.0, 1.0, 1.0) \tag{36}$$

And the exact solution at time t is provided by

$$(\rho, u, v, p)_{exact} = (1.0 + 0.2 \sin(\pi (x + y - 2t)), 1.0, 1.0, 1.0) \tag{37}$$

This problem deploys Balsara's genuinely 2D HLLE solver and terminates at $t = 1.0$. The reconstruction methods include the third-order HWENO method and the fifth-order MD_WENO_Z5 method. A CFL number of 0.6 is used. Table 2 lists the different norms and accuracy orders. We can see that both the HWENO method and the MD_WENO_Z5 method achieve their designed accuracy. Also, the absolute error values of the MD_WENO_Z5 method are less than the HWENO method.

5.2. Solution quality tests

5.2.1. Sod shock tube

The initial conditions of this problem are:

$$\begin{cases} (\rho, u, v, p)_R = (0.125, 0, 0, 0.1) & x \le 0 \\ (\rho, u, v, p)_L = (1.0, 0, 0, 1.0) & x > 0 \end{cases} \tag{38}$$

The computational domain is $[-0.5, 0.5] \times [0, 1]$, and the grid size is 200×200. The problem is stopped at the time of 0.13. The CFL number is set as 0.6. Fig. 4 shows the density distribution comparison along the line $y = 0.5$ of the HWENO and MD_WENO_Z5 methods. The numerical results of the two methods are in good agreement with the theoretical solution.

Table 2
L_1 and L_∞ errors and orders for test2 problem.

Method	Grid	L_1 error	L_1 order	L_∞ error	L_∞ order
HWENO	20×20	1.76E-04	–	3.70E-04	–
	40×40	1.43E-05	3.6299	3.01E-05	3.6171
	80×80	2.41E-06	2.5670	5.06E-06	2.5736
	160×160	3.22E-07	2.9014	6.74E-07	2.9088
	320×320	4.09E-08	2.9775	8.55E-08	2.9783
MD_WENO_Z5	20×20	2.49E-05	–	5.70E-05	–
	40×40	1.26E-06	4.3079	4.31E-06	3.7243
	80×80	4.59E-08	4.7758	2.13E-07	4.3361
	160×160	6.19E-10	6.2118	1.95E-09	6.7763
	320×320	1.93E-11	5.0042	3.37E-11	5.8509

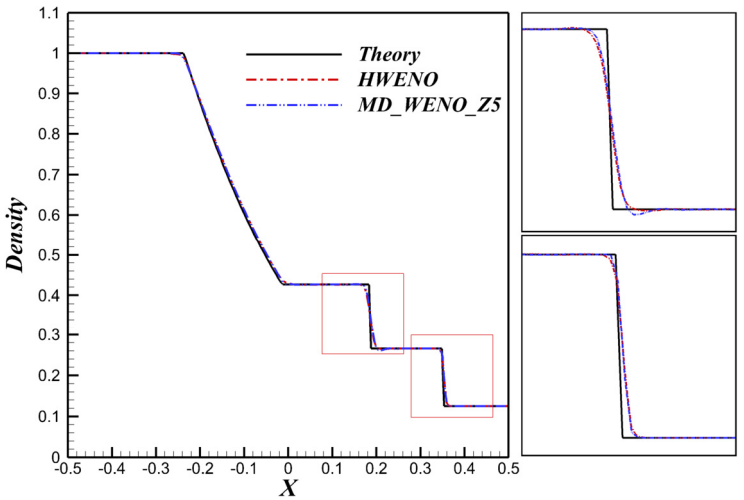


Fig. 4. Density distributions of the Sod shock tube problem, $t = 0.13$.

From the right close-up view of the shock wave and contact discontinuity in Fig. 4, it can be seen that the MD_WENO_Z5 method has a higher resolution accuracy than the HWENO method. It should be noticed that the MD_WENO_Z5 methods can theoretically achieve the fifth-order accuracy in the smooth regions, but it degrades to the third-order accuracy in the discontinuous regions. Therefore, the discontinuity resolution of the fifth-order method is close to the third-order HWENO method.

Moreover, as shown in Fig. 5, the discontinuity-detector described in this paper can accurately cover the regions where the shock wave and contact discontinuity exist in this problem, proving that the discontinuity-detector is effective.

5.2.2. Multidimensional Riemann problems

Two cases of multidimensional Riemann problems are solved to estimate the property of the fifth-order reconstruction method. The calculation is conducted on a 1000×1000 Cartesian grid on a domain of size $[0, 1] \times [0, 1]$. A CFL number of 0.6 is used for both cases. The results of two reconstruction methods, including the HWENO method and MD_WENO_Z5 method, are aligned for comparison.

Case 1. This problem describes the interaction of two shock waves and two contact discontinuities. The initial conditions are given by

$$\begin{cases} (\rho, u, v, p) = (0.5313, 0, 0, 0.4), & \text{if} \quad x > 0.5, \ y > 0.5 \\ (\rho, u, v, p) = (1.0, 0.7276, 0, 1.0), & \text{if} \quad x < 0.5, \ y > 0.5 \\ (\rho, u, v, p) = (0.8, 0, 0, 1.0), & \text{if} \quad x < 0.5, \ y < 0.5 \\ (\rho, u, v, p) = (1.0, 0, 0.7276, 1.0), & \text{if} \quad x > 0.5, \ y < 0.5 \end{cases} \tag{39}$$

Fig. 6 shows the density contours calculated by the HWENO and MD_WENO_Z5 methods for Case 1. Both reconstruction methods can accurately capture the helical flow structures at the end of the subsonic region in the center of the computational domain, and the MD_WENO_Z5 method has a higher resolution. Meanwhile, the MD_WENO_Z5 method is more accurate in capturing the two contact discontinuities, and can even describe the small vortex structures on the contact discontinuities.

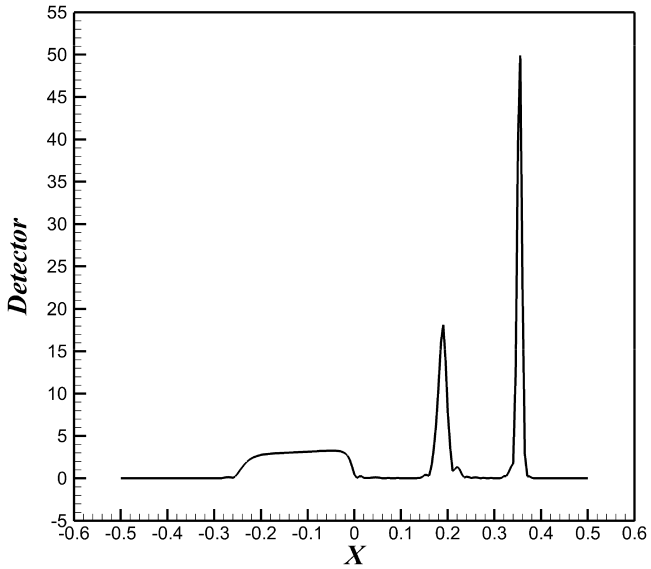
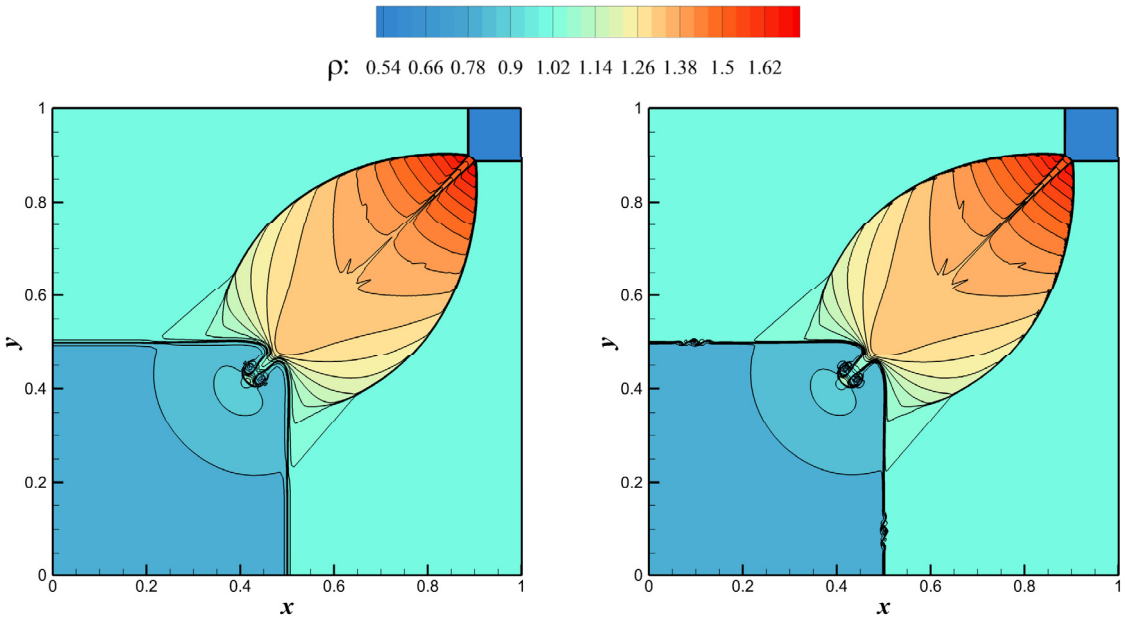


Fig. 5. Discontinuous regions of the Sod shock tube problem, $t = 0.13$.



(a) HWENO (b) MD_WENO_Z5

Fig. 6. Density contours of the two-dimensional Riemann problem: Case 1, $t = 0.25$, 1000×1000 cells, 30 equally spaced contours from 0.54 to 1.70. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Case 2. This problem describes the interaction of two contact discontinuities, one rarefaction wave and one shock wave. The initial conditions are given by

$$\begin{cases} (\rho, u, v, p) = (1.0, 0, 0.3, 1.0), & \text{if} \quad x > 0.5, \ y > 0.5 \\ (\rho, u, v, p) = (2.0, 0, -0.3, 1.0), & \text{if} \quad x < 0.5, \ y > 0.5 \\ (\rho, u, v, p) = (1.0625, 0, 0.2145, 0.4), & \text{if} \quad x < 0.5, \ y < 0.5 \\ (\rho, u, v, p) = (0.5179, 0, -0.4259, 0.4), & \text{if} \quad x > 0.5, \ y < 0.5 \end{cases} \tag{40}$$

Fig. 7 shows the density contours calculated by the HWENO and MD_WENO_Z5 methods for Case 2. The performances of the methods mentioned above are evaluated by the accuracy of capturing the two longitudinal contact discontinuities

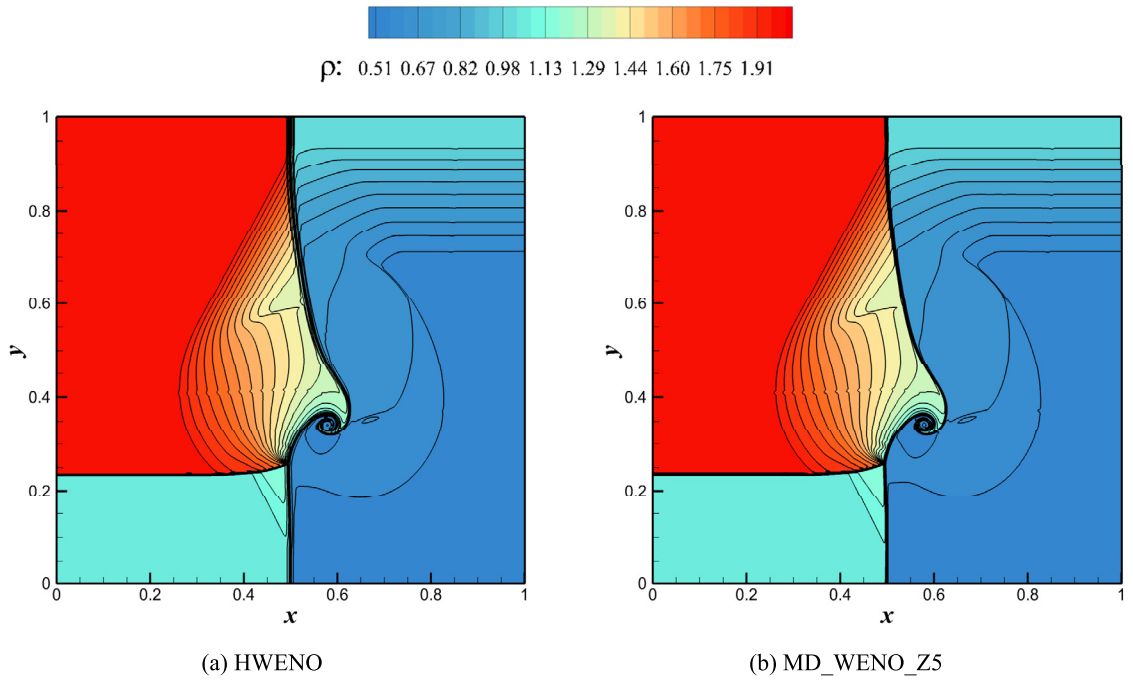


Fig. 7. Density contours of the two-dimensional Riemann problem: Case 2, $t = 0.3$, 1000×1000 cells, 30 equally spaced contours from 0.51 to 2.01.

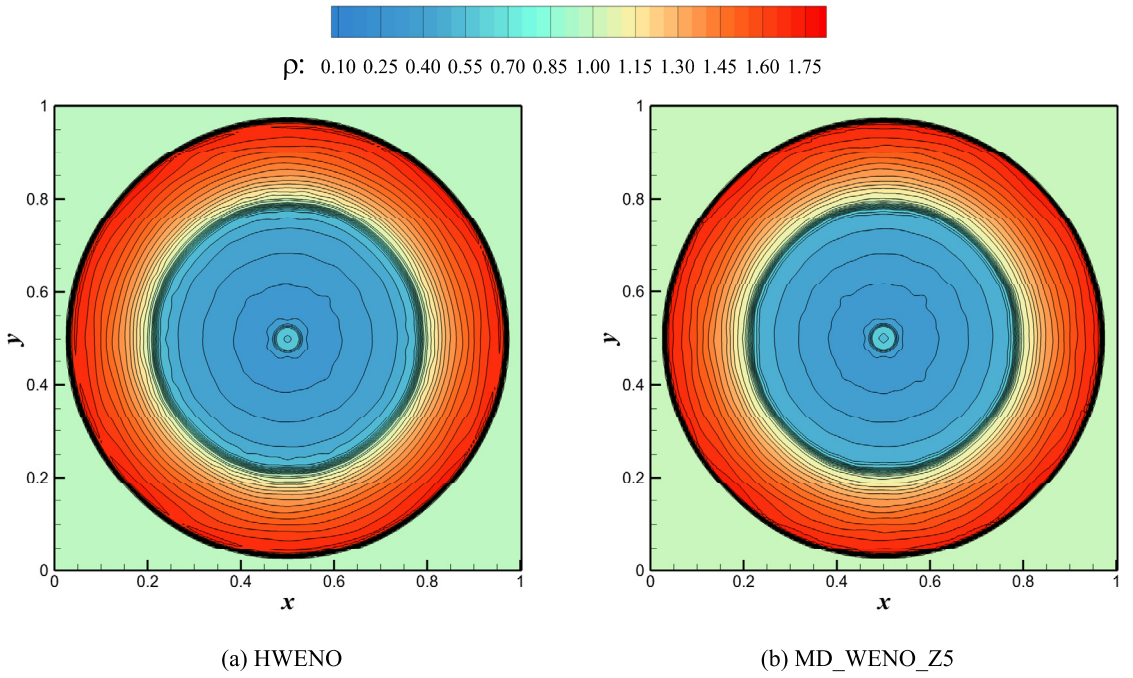
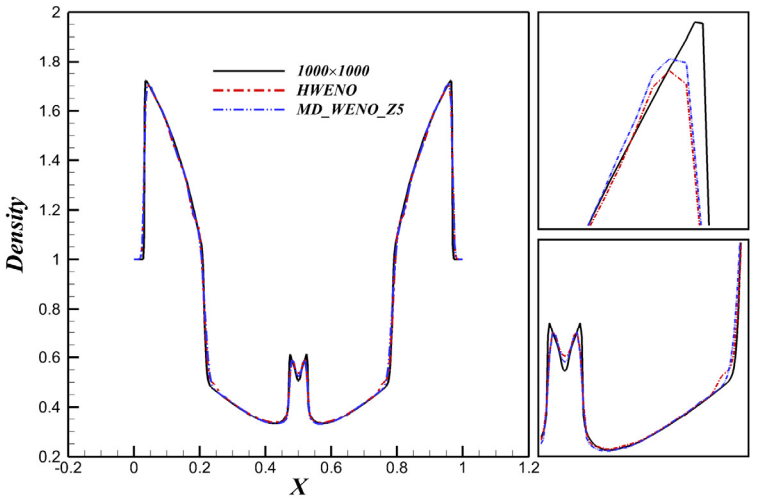


Fig. 8. Density contours of the Radial Riemann problem. $t = 0.13$, 200×200 cells, 35 equally spaced contours from 0.1 to 1.8.

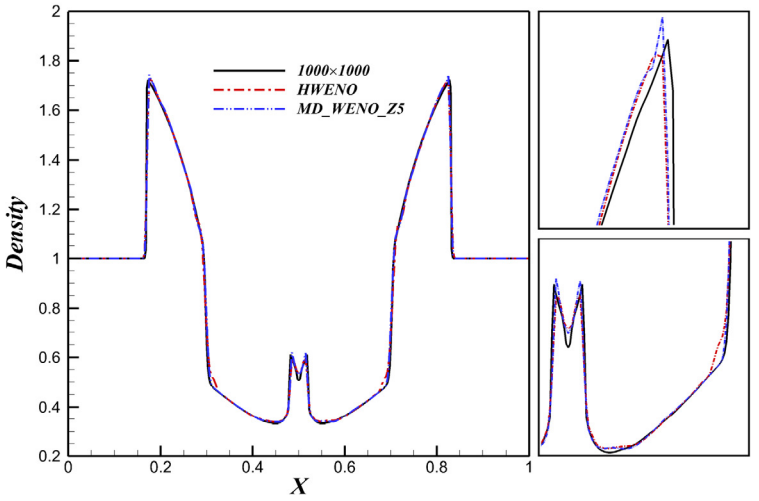
and the accuracy of capturing the helical flow structure formed at the end of the contact discontinuities. It can be seen that the fifth-order MD_WENO_Z5 method captures the two contact discontinuities and helical flow structures more accurately.

5.2.3. Radial Riemann problem

In this problem, a circular region is defined in the center of the computing domain. Inside the circle is stationary high-pressure gas and the outside is surrounded by stationary low-pressure gas. The initial conditions are given by



（a）Along the line y=0.5



(b) Along the line y=x

Fig. 9. Density distributions along different lines.

$$\begin{cases} (\rho, u, v, p) = (2.0, 0, 0, 15.0), & \text{if } r < 0.13 \\ (\rho, u, v, p) = (1.0, 0, 0, 2.0), & \text{others} \end{cases} \tag{41}$$

where $r = \sqrt{x^2 + y^2}$. The calculation is conducted on a 200×200 grid on a domain of $[0, 1] \times [0, 1]$. The CFL number is set as 0.6. The numerical results of the third-order HWENO method and fifth-order MD_WENO_Z5 method are aligned for comparison. Fig. 8 shows the density contours. These two methods can both capture the circular contact discontinuity and shock wave without oscillation essentially.

To further analyze the performances of the methods, the distribution characteristics of the flow variables along the line $y = 0.5$ and $y = x$ are extracted. Meanwhile, the numerical result of a 1000×1000 fine grid is used as the comparison benchmark. Fig. 9 (a) and (b) show the comparisons of the density distribution along the line $y = 0.5$ and $y = x$. The results of both methods are in good agreement with the benchmark. However, from the right close-up view of the contact discontinuities and shock waves, it can be seen that the result of MD_WENO_Z5 method is closer to the comparison benchmark.

Furthermore, to show the effect of our discontinuity-detector technology, the discontinuous region of the Radial Riemann problem is shown in Fig. 10. It can be seen that the detector can capture the circular shock wave and contact discontinuity accurately.

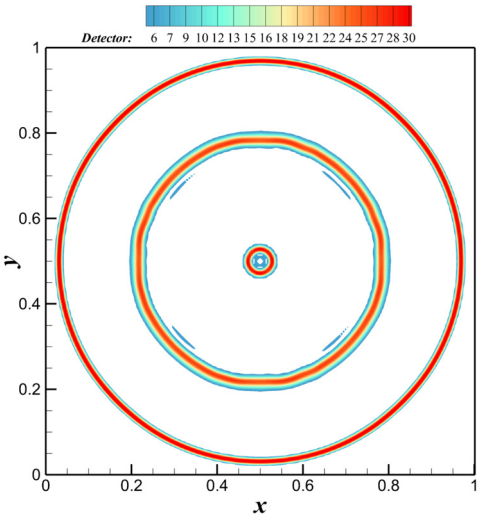


Fig. 10. Discontinuous regions of the radial Riemann problem. $t = 0.13$.

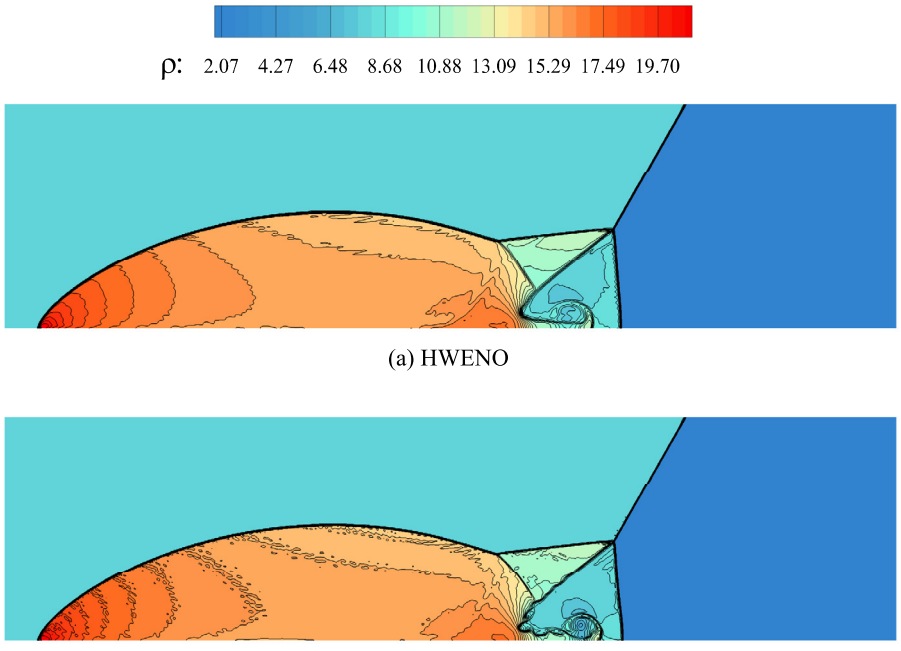


Fig. 11. Density contours of the double Mach reflection problem, 960 × 240 cells, 35 equally spaced contours from 2.07–20.8.

5.2.4. Double Mach reflection of a strong shock

This problem describes the flow evolution process after a strong shock wave with a Mach number of 10 incident to the plate at an angle of 60° at the location $x = 1/6$. The initial conditions are

$$\begin{cases} (\rho, u, v, p)_L = (8.0, 7.145, -4.125, 116.5) \\ (\rho, u, v, p)_R = (1.4, 0, 0, 1.0) \end{cases} \tag{42}$$

The calculation is conducted on a 960 × 240 grid on a domain of $[0, 4] \times [0, 1]$. The CFL number is set as 0.6. Fig. 11 shows the density contours calculated by the HWENO and MD_WENO_Z5 methods at time $t = 0.2$. Fig. 12 shows a close-up view of the double Mach reflection region. We can see that the fifth-order MD_WENO_Z5 method capture the contact discontinuity more clearly and depicts finer vortical structure near the Mach stem compared to the third-order method. It can be argued that the fifth-order MD_WENO_Z5 method obtains smaller dissipation and higher accuracy. In addition,

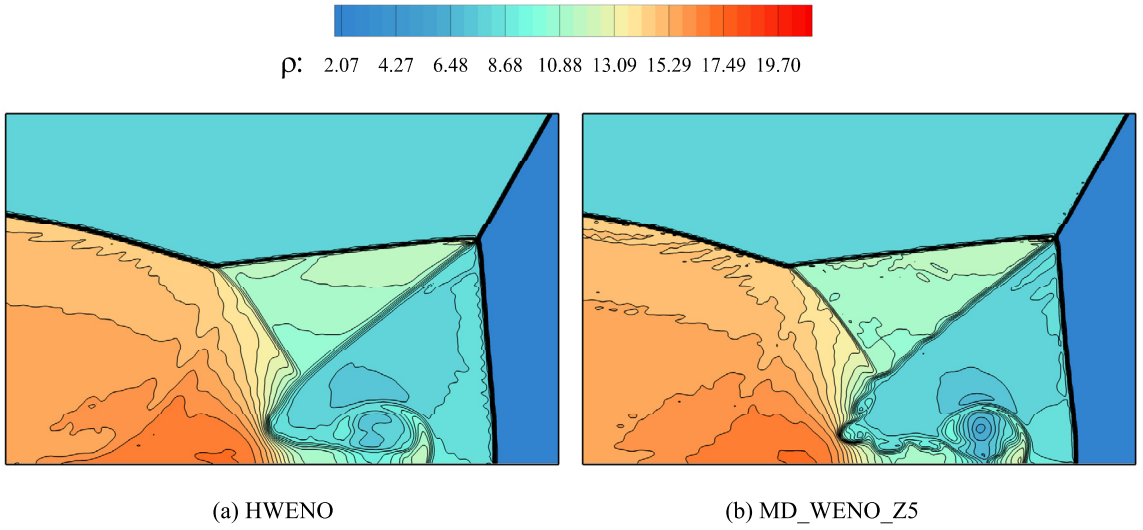


Fig. 12. Close-up view of the double Mach reflection region, 960 × 240 cells, 35 equally spaced contours from 2.07–20.8.

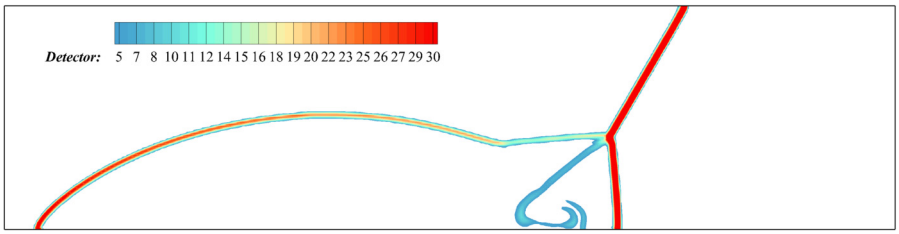


Fig. 13. Discontinuous regions of the double Mach reflection problem. $t = 0.2$.

we display the discontinuous regions captured by the discontinuity-detector in Fig. 13, which covers the shock waves and contact discontinuities clearly and proves the effectiveness of the discontinuity-detector in this paper.

6. Conclusions

In this study, a multidimensional fifth-order reconstruction method denoted by MD_WENO_Z5 is developed for Balsara's genuinely 2D HLLE Riemann solver. The critical idea of the reconstruction method is to utilize the general WENO reconstruction formula to complete the reconstruction procedure along two spatial directions in turns, and use the Gaussian quadrature approach to achieve the fifth-order accuracy for the numerical flux integration at the interfaces. Four input states around each Gaussian point are obtained by the reconstruction formula, and the 2D Riemann problem is defined and solved in these Gaussian points. Due to the fact that the fifth-order reconstruction method used in complex flows may cause overshoots to reduce the computational stability, we develop a new limiter to enhance the robustness of the scheme. This limiter adopts the discontinuity-detector technology based on the standard deviation coefficient, and preserves accuracy in the smooth regions. Solution accuracy tests indicate that the fifth-order method can achieve its design accuracy. The other test cases show that the developed multidimensional fifth-order scheme performs well and can accurately capture the complex flow structures. Besides, the discontinuity-detector technology proposed in this study can depict the discontinuity accurately.

CRediT authorship contribution statement

Boxiao Zhou: Validation, Data Curation, Writing – Original Draft. **Feng Qu:** Conceptualization, Methodology, Software, Writing – Review & Editing, Supervision, Project administration, Funding acquisition. **Qingsong Liu:** Formal analysis, Investigation, Funding acquisition. **Di Sun:** Writing – Review. **Junqiang Bai:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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